Lecture L02
Exponential Families and Sufficient Statistics

A common representation for probabilistic models

Probabilistic Machine Learning 2027-1

YOUR NOTES

Section 1

The exponential family

Features, matching coefficients and normalization

YOUR NOTES

Lecture L02 — Exponential Families

Three components define the exponential family

The exponential family is a family of distributions that have some characteristics in common

- A small set of features $T_j(x)$ define the likelihood.
  - Not the entire dataset
- The distribution can be parameterized with natural parameters η_j
  - Each feature's contribution $T_j(x)$ to the likelihood depends on its natural parameter.
- The likelihood factors: $p_\eta(x) \propto h(x)\exp(\eta^T T(x))$

$$p_\eta(x) = \frac{h(x)\exp(\eta^T T(x))}{Z(\eta)} = h(x)\exp(\eta^T T(x) - A(\eta))$$

$$A(\eta) = \log Z(\eta)$$

1. Common representation

S1.02 · Slide 3/67

STUDY NOTE

The common form separates observation features from the coefficients that weight them and the factor that normalizes the distribution. Identifying these roles lets us transfer one learning argument between distributions rather than derive a separate estimation method for each example.

YOUR NOTES

Lecture L02 — Exponential Families

Algebra rewrites Bernoulli in exponential form

Carried forward Probability background · Bernoulli; S1.02 · Exponential family form

Bernoulli distribution: $p(x|\theta) = \theta^x(1-\theta)^{1-x}, \quad x \in \{0,1\}, \quad 0 < \theta < 1$

Exponential family form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

$$\begin{aligned} p(x|\theta) &= \theta^x(1-\theta)^{1-x} \\ &= \exp(\log(\theta^x(1-\theta)^{1-x})) \\ &= \exp(x\log\theta + (1-x)\log(1-\theta)) \end{aligned}$$

Collect x

$$p(x|\theta) = \exp\left(x\log\left(\frac{\theta}{1-\theta}\right) + \log(1-\theta)\right)$$

- $T(x) = x$ is the observation **statistic**.
  $\eta = \log\left(\frac{\theta}{1-\theta}\right)$ is the **natural coefficient**.
  $h(x) = 1$ is the fixed **base factor**.
  Remainder: $\log(1-\theta)$, still in θ.

1. Common representation

S1.03 · Slide 4/67

STUDY NOTE

Taking logarithms exposes the coefficient of the binary observation. The resulting log odds is a new coordinate for the same Bernoulli probabilities; the remaining term must still be expressed in that coordinate to complete the exponential-family representation.

YOUR NOTES

Lecture L02 — Exponential Families

Put the log normalizer in terms of η

Carried forward S1.03 · Identified coordinate and remainder

$\eta = \log\left(\frac{\theta}{1-\theta}\right)$, $\log(1-\theta)$ is the remainder, $0 < \theta < 1$

$p(x|\theta) = \theta^x(1-\theta)^{1-x}$ Exponential form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

$$\eta = \log\left(\frac{\theta}{1-\theta}\right) \Rightarrow e^\eta = \frac{\theta}{1-\theta} \Rightarrow e^\eta(1-\theta) = \theta \Rightarrow e^\eta = (1+e^\eta)\theta$$

$$\theta(\eta) = \frac{e^\eta}{1+e^\eta} \Rightarrow 1-\theta(\eta) = 1 - \frac{e^\eta}{1+e^\eta} = \frac{1}{1+e^\eta}$$

Substitute the inverse for θ in the remainder

$$A(\eta) = -\log(1-\theta(\eta)) = -\log\left(\frac{1}{1+e^\eta}\right) = \log(1+e^\eta)$$

$$p_\eta(x) = p(x|\theta(\eta)) = \exp(\eta x - A(\eta))$$

1. Common representation

S1.04 · Slide 5/67

STUDY NOTE

Solving the log-odds relation gives the probability as a function of the natural parameter. Substituting that inverse into the remainder expresses every parameter-dependent term in the new coordinate and completes the Bernoulli match.

YOUR NOTES

Lecture L02 — Exponential Families

The partition function normalizes the exponential form

Carried forward S1.02 · Z and A forms; S1.04 · Bernoulli specialization

$p_\eta(x) = \exp(\eta x - A(\eta)) = \frac{e^{\eta x}}{Z(\eta)}, \quad Z(\eta) = e^{A(\eta)} > 0, \quad x \in \{0, 1\}$

$$\sum_x \frac{e^{\eta x}}{Z(\eta)} = \frac{1}{Z(\eta)} \sum_x e^{\eta x} = 1$$

$$Z(\eta) = \sum_x e^{\eta x} = 1 + e^{\eta}$$

- **Partition function**: sum or integral of unnormalized weights.
  Dividing by $Z(\eta)$ gives total probability 1.
  Bayesian **evidence** similarly normalizes a posterior over unknowns rather than outcomes.

1. Common representation

S1.05 · Slide 6/67

STUDY NOTE

Summing the two Bernoulli outcome weights determines the divisor that makes their probabilities total one. This gives a concrete meaning to the partition function and prepares us to use the same normalization principle for more general supports.

YOUR NOTES

Lecture L02 — Exponential Families

The natural coordinate is the log ratio of the two outcome probabilities

Carried forward S1.05 · Bernoulli exponential ratio
$p_\eta(x) = \frac{e^{\eta x}}{1+e^\eta}, \quad x \in \{0, 1\}$

Substitute $x = 0$ and $x = 1$ into the ratio:

$$p_\eta(0) = \frac{1}{1+e^\eta}, \quad p_\eta(1) = \frac{e^\eta}{1+e^\eta}$$

$$\frac{p_\eta(1)}{p_\eta(0)} = e^\eta \Rightarrow \eta = \log\left(\frac{p_\eta(1)}{p_\eta(0)}\right)$$

$\eta < 0 \Rightarrow p_\eta(1) < p_\eta(0)$
$\eta = 0 \Rightarrow p_\eta(1) = p_\eta(0) = \frac{1}{2}$
$\eta > 0 \Rightarrow p_\eta(1) > p_\eta(0)$

$p_\eta(0)$	$p_\eta(1)$	η (approx.)
0.99	0.01	−4.595
0.90	0.10	−2.197
0.50	0.50	0
0.10	0.90	2.197
0.01	0.99	4.595

- Adding 1 to η multiplies the odds by e.
- $\eta \to -\infty$ gives $p_\eta(1) \to 0$; $\eta \to +\infty$ gives $p_\eta(1) \to 1$.

1. Common representation

S1.06 · Slide 7/67

STUDY NOTE

The natural parameter measures relative preference between the two outcomes, whereas the probability is restricted to the interval from zero to one. This distinction makes the coefficient interpretable and explains why boundary probabilities arise only as infinite log-odds limits.

YOUR NOTES

Lecture L02 — Exponential Families

Section 1 summary: Statistics, coefficients and normalization

$$\theta^x(1-\theta)^{1-x} = \exp(\eta x - A(\eta))$$
$$T(x) = x, \quad \eta = \log\left(\tfrac{\theta}{1-\theta}\right), \quad h(x) = 1$$
$$Z(\eta) = 1 + e^{\eta}, \quad A(\eta) = \log Z(\eta), \quad e^{-A} = \tfrac{1}{Z}$$

TAKE HOME

Bernoulli's statistic is x, its coefficient is log odds, and division by $1 + e^{\eta}$ recovers the same probabilities.
The exponential family form separates features, natural coefficients and normalization.

1. Common representation

S1.07 · Slide 8/67

STUDY NOTE

The Bernoulli rewrite preserves the original distribution while separating its observation statistic, natural coefficient and normalizer. These are the pieces to identify when testing whether a different model has the same exponential-family structure.

YOUR NOTES

Section 2

The Gaussian distribution is also an exponential family

Rewrite the familiar density in terms of $h(x)$, $T(x)$, η, and $A(\eta)$

YOUR NOTES

Lecture L02 — Exponential Families

Our goal: the Gaussian in exponential family form

Carried forward L01 · Gaussian; S1 · EF form

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

To derive

$$p_\eta(x) = (2\pi)^{-\frac{D}{2}}\exp(\eta^T T(x) - A(\eta))$$

$$\eta = \begin{pmatrix} \Lambda\mu \\ -\frac{1}{2}\,\mathrm{vec}(\Lambda) \end{pmatrix}, \quad T(x) = \begin{pmatrix} x \\ \mathrm{vec}(xx^T) \end{pmatrix}$$

Precision matrix: $\Lambda \triangleq \Sigma^{-1}$.

$$h(x) = (2\pi)^{-\frac{D}{2}}, \quad A(\eta) = -\frac{1}{2}\log|\Lambda| + \frac{1}{2}\mu^T\Lambda\mu$$

Column stacking: stack columns from left to right.

$$B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$\mathrm{vec}(B) = (a, c, b, d)^T$

2. Gaussian representation

S2.02 · Slide 10/67

STUDY NOTE

The Gaussian requires both linear and quadratic observation features, so its natural parameter is a vector of matching coefficients. Precision and column stacking provide the notation needed to recognize these pieces in the familiar density without changing the distribution.

YOUR NOTES

Lecture L02 — Exponential Families

Replace covariance by precision, then take the logarithm

Carried forward L01 · Gaussian; S1 · EF; S2.02 · Precision

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Use: $\Lambda = \Sigma^{-1}, \quad x, \mu \in \mathbb{R}^D, \quad \Sigma, \Lambda \in \mathbb{R}^{D\times D}$

Current development

$$\Sigma = \Sigma^T \succ 0 \Rightarrow \Lambda = \Lambda^T \succ 0, \quad |\Sigma|^{-\frac{1}{2}} = |\Lambda|^{\frac{1}{2}}$$

$$p(x) = (2\pi)^{-\frac{D}{2}}|\Lambda|^{\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Lambda(x-\mu)\right)$$

$$\log p(x) = -\frac{D}{2}\log(2\pi) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}(x-\mu)^T\Lambda(x-\mu)$$

STUDY NOTE

Using inverse covariance simplifies the quadratic expression, and taking the logarithm turns products into additive terms that can be matched separately. The positive-definite covariance assumption makes the precision and determinant manipulations valid.

YOUR NOTES

Lecture L02 — Exponential Families

Expand the quadratic inside the current log density

Carried forward S2.03 · Complete log density; L01/S1 · Reference forms

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Current: $\log p(x) = -\frac{D}{2}\log(2\pi) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}(x-\mu)^T\Lambda(x-\mu)$

Use: $x, \mu \in \mathbb{R}^D, \quad \Lambda = \Lambda^T \in \mathbb{R}^{D\times D}$

Updated: $$\log p(x) = -\frac{D}{2}\log(2\pi) + (\Lambda\mu)^T x - \frac{1}{2}x^T\Lambda x + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\mu^T\Lambda\mu$$

Expand the quadratic term

$$(x-\mu)^T\Lambda(x-\mu)$$
$$= x^T\Lambda x - x^T\Lambda\mu - \mu^T\Lambda x + \mu^T\Lambda\mu$$
$$x^T\Lambda\mu = \left(x^T\Lambda\mu\right)^T = \mu^T\Lambda^T x = \mu^T\Lambda x = (\Lambda\mu)^T x$$

2. Gaussian representation

S2.04 · Slide 12/67

STUDY NOTE

Expanding the centered quadratic separates terms linear in the observation, terms quadratic in it and terms independent of it. Symmetry of the precision matrix combines the two cross terms, revealing the coefficient of the linear feature block.

YOUR NOTES

Lecture L02 — Exponential Families

Rewrite the density's quadratic term as a feature dot product

Carried forward S2.04 · Updated log density; L01/S1 · Reference forms

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Current: $\log p(x) = -\frac{D}{2}\log(2\pi) + (\Lambda\mu)^T x - \frac{1}{2}x^T\Lambda x + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\mu^T\Lambda\mu$

Updated: $$\log p(x) = -\frac{D}{2}\log(2\pi) + (\Lambda\mu)^T x - \frac{1}{2}\mathrm{vec}(\Lambda)^T\,\mathrm{vec}(xx^T) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\mu^T\Lambda\mu$$

Rewrite the quadratic term · D^2 matching entries

$$x^T\Lambda x = \sum_{i=1}^{D}\sum_{j=1}^{D}\Lambda_{ij}x_i x_j$$

$$= \sum_{i,j}\Lambda_{ij}\left(xx^T\right)_{ij} = \mathrm{vec}(\Lambda)^T\,\mathrm{vec}\left(xx^T\right)$$

STUDY NOTE

The quadratic form sums products of corresponding entries from the precision matrix and the observation outer product. Stacking both matrices in the same column order turns this sum into a dot product, making it compatible with the exponential-family pairing.

YOUR NOTES

Lecture L02 — Exponential Families

Match the density's linear and quadratic coefficient–feature blocks

Carried forward S2.05 · Vectorized log density; L01/S1 · Reference forms

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Current: $\log p(x) = -\frac{D}{2}\log(2\pi) + (\Lambda\mu)^T x - \frac{1}{2}\mathrm{vec}(\Lambda)^T\,\mathrm{vec}(xx^T) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\mu^T\Lambda\mu$

Updated: $$\log p(x) = \eta^T T(x) - \frac{D}{2}\log(2\pi) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\mu^T\Lambda\mu$$

Pair the linear and quadratic coefficient–feature blocks

$$\eta \triangleq \begin{pmatrix} \Lambda\mu \\ -\frac{1}{2}\,\mathrm{vec}(\Lambda) \end{pmatrix}, \quad T(x) \triangleq \begin{pmatrix} x \\ \mathrm{vec}(xx^T) \end{pmatrix}$$

$$\eta^T T(x) = (\Lambda\mu)^T x + \left[-\frac{1}{2}\,\mathrm{vec}(\Lambda)\right]^T \mathrm{vec}(xx^T)$$

2. Gaussian representation

S2.06 · Slide 14/67

STUDY NOTE

The linear and quadratic contributions can now be assembled into one parameter vector and one feature vector. Their inner product replaces exactly the observation-dependent terms in the log density, leaving the base and normalizer to be identified.

YOUR NOTES

Lecture L02 — Exponential Families

Write the natural parameter and density using the precision-weighted mean

Carried forward S2.06 · Matched density and block definitions; L01/S1 · Reference forms

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Current: $p(x) = (2\pi)^{-\frac{D}{2}}\exp\left(\eta^T T(x) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\mu^T\Lambda\mu\right)$

Use: $\eta = \begin{pmatrix}\Lambda\mu \\ -\frac{1}{2}\mathrm{vec}(\Lambda)\end{pmatrix}$, $T(x) = \begin{pmatrix}x \\ \mathrm{vec}(xx^T)\end{pmatrix}$

Updated: $$\log p(x) = \eta^T T(x) - \frac{D}{2}\log(2\pi) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\xi^T\Lambda^{-1}\xi$$

Use $\xi \triangleq \Lambda\mu$, with $\Lambda = \Lambda^T \succ 0$

$\xi = \Lambda\mu \Rightarrow \mu = \Lambda^{-1}\xi$

$\eta = \begin{pmatrix}\xi \\ -\frac{1}{2}\mathrm{vec}(\Lambda)\end{pmatrix}$

$\mu^T\Lambda\mu = (\Lambda^{-1}\xi)^T\Lambda(\Lambda^{-1}\xi) = \xi^T\Lambda^{-1}\xi$

2. Gaussian representation

S2.07 · Slide 15/67

STUDY NOTE

The precision-weighted mean is the linear natural-parameter block. Rewriting the remaining mean quadratic in terms of this block ensures that the eventual normalizer is expressed through the natural coordinates rather than a mixture of old and new parameters.

YOUR NOTES

Lecture L02 — Exponential Families

Match the remaining log-density terms to h and A

Carried forward S2.07 · Natural-coordinate density and updated blocks; L01/S1 · Reference forms

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Current: $\log p(x) = \eta^T T(x) - \frac{D}{2}\log(2\pi) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\xi^T\Lambda^{-1}\xi$

Use: $\eta = \begin{pmatrix}\xi \\ -\frac{1}{2}\mathrm{vec}(\Lambda)\end{pmatrix}$, $T(x) = \begin{pmatrix}x \\ \mathrm{vec}(xx^T)\end{pmatrix}$

Updated: $\log p(x) = \log h(x) + \eta^T T(x) - A(\eta)$

Match the non-data terms in $\log p = \log h + \eta^T T - A$

$$\log h(x) = -\frac{D}{2}\log(2\pi) \Rightarrow h(x) = (2\pi)^{-\frac{D}{2}}$$

$$-A(\eta) = \frac{1}{2}\log|\Lambda| - \frac{1}{2}\xi^T\Lambda^{-1}\xi \Rightarrow A(\eta) = -\frac{1}{2}\log|\Lambda| + \frac{1}{2}\xi^T\Lambda^{-1}\xi$$

STUDY NOTE

After extracting the feature pairing, the fixed factor belongs in the base and the parameter-only remainder is minus the log normalizer. Tracking that minus sign completes the Gaussian match and determines its normalizer without a second integration calculation.

YOUR NOTES

Lecture L02 — Exponential Families

The completed density is the original Gaussian in new coordinates

Carried forward S2.07–08 · Natural-coordinate log density and scalar components; L01/S1 · Reference forms

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Current: $\log p(x) = \eta^T T(x) - \frac{D}{2}\log(2\pi) + \frac{1}{2}\log|\Lambda| - \frac{1}{2}\xi^T\Lambda^{-1}\xi$

Use: $h(x) = (2\pi)^{-\frac{D}{2}}, \quad A(\eta) = -\frac{1}{2}\log|\Lambda| + \frac{1}{2}\xi^T\Lambda^{-1}\xi$

Updated: $p(x) = (2\pi)^{-\frac{D}{2}}\exp(\eta^T T(x) - A(\eta))$

Exponentiate the completed log-density match

$$\log p(x) = \log h(x) + \eta^T T(x) - A(\eta)$$

$$p(x) = h(x)\exp(\eta^T T(x) - A(\eta))$$

2. Gaussian representation

S2.09 · Slide 17/67

STUDY NOTE

Exponentiating the completed log-density match returns the whole normalized Gaussian, not merely a proportional kernel. The calculation establishes a new parameterization of a familiar distribution, which allows the later exponential-family identities to apply to it.

YOUR NOTES

Lecture L02 — Exponential Families

The Gaussian exponential family: all components in final form

Original

$$p(x) = (2\pi)^{-\frac{D}{2}} |\Sigma|^{-\frac{1}{2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

Exponential family form

$$p_\eta(x) = h(x) \exp(\eta^T T(x) - A(\eta))$$

$$\eta = \begin{pmatrix} \xi \\ -\frac{1}{2}\,\mathrm{vec}(\Lambda) \end{pmatrix}, \quad T(x) = \begin{pmatrix} x \\ \mathrm{vec}(xx^T) \end{pmatrix}$$

$$\Lambda = \Sigma^{-1}, \quad \xi = \Lambda\mu$$

$$h(x) = (2\pi)^{-\frac{D}{2}}, \quad A(\eta) = -\frac{1}{2}\log|\Lambda| + \frac{1}{2}\xi^T \Lambda^{-1}\xi$$

2. Gaussian representation

S2.10 · Slide 18/67

STUDY NOTE

The original density and its exponential-family form describe the same model. Keeping the parameter blocks, feature blocks, base and normalizer together provides the reference needed to carry out a concrete substitution correctly.

YOUR NOTES

Lecture L02 — Exponential Families

Expand the squares inside the complete two-dimensional density

Carried forward S2.03/10 · Gaussian precision density and EF form

$p(x) = (2\pi)^{-\frac{D}{2}}|\Lambda|^{\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Lambda(x-\mu)\right)$

$p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Updated: $$p(x) = \frac{1}{2\pi}\exp\left(x_1 + 2x_2 - \frac{1}{2}x_1^2 - \frac{1}{2}x_2^2 - \frac{5}{2}\right)$$

Choose the example, substitute, then expand the squares

$$D = 2, \quad \mu = (1,2)^T, \quad \Sigma = \Lambda = I_2$$

$$p(x) = \frac{1}{2\pi}\exp\left(-\frac{1}{2}\left[(x_1-1)^2 + (x_2-2)^2\right]\right)$$

$$p(x) = \frac{1}{2\pi}\exp\left(-\frac{1}{2}\left[x_1^2 - 2x_1 + 1 + x_2^2 - 4x_2 + 4\right]\right)$$

STUDY NOTE

Choosing a mean and identity covariance makes the general Gaussian mapping inspectable in two dimensions. Expanding inside the complete density keeps its normalization and constant term visible while exposing the linear and quadratic observation terms.

YOUR NOTES

Lecture L02 — Exponential Families

Build the coefficient and feature vectors from their blocks

Carried forward S2.10–11 · Block definitions and expanded example

$p(x) = \frac{1}{2\pi}\exp\left(x_1 + 2x_2 - \frac{1}{2}x_1^2 - \frac{1}{2}x_2^2 - \frac{5}{2}\right)$

$\eta = \begin{pmatrix} \xi \\ -\frac{1}{2}\,\mathrm{vec}(\Lambda) \end{pmatrix}, \quad T(x) = \begin{pmatrix} x \\ \mathrm{vec}(xx^T) \end{pmatrix}, \quad \xi = \Lambda\mu$

$\Lambda = I_2, \quad \mu = (1,2)^T$

Updated:

$$\eta = \left(1, 2, -\frac{1}{2}, 0, 0, -\frac{1}{2}\right)^T \qquad T(x) = (x_1, x_2, x_1^2, x_1x_2, x_1x_2, x_2^2)^T$$

Compute the two coefficient blocks and the quadratic features

$\mathrm{vec}(\Lambda) = (1,0,0,1)^T$

$\xi = \Lambda\mu = (1,2)^T$

$\mathrm{vec}(xx^T) = (x_1^2, x_1x_2, x_1x_2, x_2^2)^T$

STUDY NOTE

The coefficient and feature vectors are assembled using the same column-stacking convention as the general derivation. Repeated off-diagonal features reflect the full matrix representation; their matching coefficients determine their actual contribution to this example.

YOUR NOTES

Lecture L02 — Exponential Families

Apply the computed blocks and return to the complete density

Carried forward S2.11–12 · Expanded density and computed vectors

$p(x) = \frac{1}{2\pi}\exp\left(x_1 + 2x_2 - \frac{1}{2}x_1^2 - \frac{1}{2}x_2^2 - \frac{5}{2}\right)$

$\eta = \left(1, 2, -\frac{1}{2}, 0, 0, -\frac{1}{2}\right)^T$

$T(x) = \left(x_1, x_2, x_1^2, x_1x_2, x_1x_2, x_2^2\right)^T$

Updated: $p(x) = \dfrac{1}{2\pi}\exp\left(\eta^T T(x) - \dfrac{5}{2}\right)$

Multiply the carried vectors, then replace the data terms

$$\eta^T T(x) = (1,2)(x_1,x_2)^T - \frac{1}{2}(1,0,0,1)\left(x_1^2, x_1x_2, x_1x_2, x_2^2\right)^T$$

$$\eta^T T(x) = x_1 + 2x_2 - \frac{1}{2}x_1^2 - \frac{1}{2}x_2^2$$

2. Gaussian representation

S2.13 · Slide 21/67

STUDY NOTE

Multiplying the assembled vectors recovers exactly the observation-dependent part of the expanded density. Substituting that inner product back into the whole expression checks the representation while keeping the parameter-only constant available for the final match.

YOUR NOTES

Lecture L02 — Exponential Families

Identify h and A in the two-dimensional density

Carried forward S2.08/10/12–13 · Gaussian components and paired example

$p(x) = \frac{1}{2\pi}\exp\left(\eta^T T(x) - \frac{5}{2}\right)$

$p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

$h(x) = (2\pi)^{-\frac{D}{2}}, \quad A(\eta) = -\frac{1}{2}\log|\Lambda| + \frac{1}{2}\xi^T\Lambda^{-1}\xi$

$D = 2, \quad \Lambda = I_2, \quad \xi = (1,2)^T$

Updated: $$p(x) = h(x)\exp(\eta^T T(x) - A(\eta)), \quad h(x) = \frac{1}{2\pi}, \quad A(\eta) = \frac{5}{2}$$

Substitute the chosen parameters into the known component formulas

$$A(\eta) = -\frac{1}{2}\log|I_2| + \frac{1}{2}(1,2)I_2^{-1}(1,2)^T$$

$$A(\eta) = -\frac{1}{2}\log 1 + \frac{1}{2}(1^2 + 2^2) = \frac{5}{2}, \quad h(x) = (2\pi)^{-\frac{2}{2}} = \frac{1}{2\pi}$$

2. Gaussian representation

S2.14 · Slide 22/67

STUDY NOTE

Substitution into the general Gaussian component formulas gives the same constant obtained by expanding the squares. This numerical match checks both the normalizer sign and the separation between the fixed base and the natural-parameter contribution.

YOUR NOTES

Lecture L02 — Exponential Families

Section 2 summary: The same Gaussian in natural coordinates

Carried forward L01 · Gaussian; S1 · EF form

Original: $p(x) = (2\pi)^{-\frac{D}{2}}|\Sigma|^{-\frac{1}{2}}\exp\left(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\right)$

EF form: $p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

$$\eta = \begin{pmatrix} \xi \\ -\frac{1}{2}\operatorname{vec}(\Lambda) \end{pmatrix}, \quad T(x) = \begin{pmatrix} x \\ \operatorname{vec}(xx^T) \end{pmatrix}, \quad \xi = \Lambda\mu$$

$$h(x) = (2\pi)^{-\frac{D}{2}}, \quad A(\eta) = -\frac{1}{2}\log|\Lambda| + \frac{1}{2}\xi^T\Lambda^{-1}\xi$$

$$p(x) = h(x)\exp(\eta^T T(x) - A(\eta))$$

TAKE HOME Linear and quadratic observation features pair with precision-weighted mean and precision coefficients. The base and log normalizer complete the original normalized Gaussian; the full vectorization is nonminimal.

2. Gaussian representation

S2.15 · Slide 23/67

STUDY NOTE

The Gaussian has the same exponential-family structure as Bernoulli, with vector and quadratic features replacing a single binary statistic. Its full vectorization repeats some features, so this convenient representation contains more coordinates than are independently necessary.

YOUR NOTES

Section 3

Normalization determines the moments of the statistics

Expected features, covariance and curvature

YOUR NOTES

Lecture L02 — Exponential Families

The normalizer sums or integrates the outcome weights

Carried forward S1–S2 · Exponential family form
$p_\eta(x) = h(x)e^{\eta^T T(x) - A(\eta)}, \quad A(\eta) = \log Z(\eta)$

$$Z(\eta) \triangleq \int_{\mathcal{X}} h(x)e^{\eta^T T(x)}\, \mathrm{d}\nu(x)$$

$$0 < Z(\eta) < \infty, \quad T(x) = (T_1(x), ..., T_d(x))^T$$

- Fix the support $\mathcal{X}$, base $h \geq 0$ and features T while η varies.
- ν is counting measure for discrete outcomes, or Lebesgue measure for densities.

STUDY NOTE

The normalizer accounts for all possible outcomes while the support, base and features stay fixed. Requiring a finite positive value identifies where this representation defines a probability distribution and supplies the starting point for extracting model moments.

YOUR NOTES

Lecture L02 — Exponential Families

Differentiate the logarithm and the normalizer

Carried forward S3.02 · EF density and normalizer

$p_\eta(x) = h(x)e^{\eta^T T(x) - A(\eta)}, \quad A(\eta) = \log Z(\eta)$

$Z(\eta) = \int_{\mathcal{X}} h(x)e^{\eta^T T(x)}\, \mathrm{d}\nu(x)$

Updated: $$\frac{\partial A(\eta)}{\partial \eta_j} = \frac{1}{Z(\eta)} \int T_j(x) h(x) e^{\eta^T T(x)}\, \mathrm{d}\nu(x)$$

Differentiate the log; assume differentiation may pass through the integral

$$\frac{\partial A}{\partial \eta_j} = \frac{1}{Z}\frac{\partial Z}{\partial \eta_j} = \frac{1}{Z}\int h(x)e^{\eta^T T(x)} \frac{\partial(\eta^T T(x))}{\partial \eta_j}\, \mathrm{d}\nu(x)$$

- $\frac{\partial(\eta^T T(x))}{\partial \eta_j} = T_j(x).$

3. Normalizer and moments

S3.03 · Slide 26/67

STUDY NOTE

The logarithm contributes a reciprocal normalizer, while differentiating the exponential introduces the selected feature. Under the stated interchange assumption, these two elementary derivatives produce an integral that can be recognized as a model expectation.

YOUR NOTES

Lecture L02 — Exponential Families

Dividing by the normalizer reveals an expectation

Carried forward S3.03 · Updated derivative; S1–S2 · EF density

$\partial_j A = \frac{1}{Z}\int T_j(x)h(x)e^{\eta^T T(x)}\,\mathrm{d}\nu(x), \quad \partial_j \triangleq \frac{\partial}{\partial \eta_j}$

$p_\eta(x) = h(x)e^{\eta^T T(x) - A(\eta)}, \quad \frac{1}{Z} = e^{-A(\eta)}$

Updated: $\nabla A(\eta) = E_\eta[T(X)]$

Bring the x-independent factor $\frac{1}{Z} = e^{-A}$ into the integral

$$\partial_j A = \int T_j(x)h(x)e^{\eta^T T(x) - A(\eta)}\,\mathrm{d}\nu(x)$$

$$= \int T_j(x)p_\eta(x)\,\mathrm{d}\nu(x) = E_\eta\left[T_j(X)\right]$$

3. Normalizer and moments

S3.04 · Slide 27/67

STUDY NOTE

Moving the reciprocal normalizer inside the integral turns the unnormalized weight into the model density. Each derivative of A is therefore the expected value of its corresponding feature, connecting a normalization calculation to a probabilistic quantity.

YOUR NOTES

Lecture L02 — Exponential Families

η contains coefficients; $m(\eta)$ contains predicted averages

Carried forward S3.04 · First-moment identity
$\nabla A(\eta) = E_\eta[T(X)]$

$$\eta \quad \to \quad p_\eta \quad \to \quad m(\eta) \triangleq E_\eta[T(X)]$$

Bernoulli: $\eta \in \mathbb{R} \quad \to \quad m(\eta) = \theta = \dfrac{e^\eta}{1+e^\eta} \in (0,1)$

- η_j is the coefficient of $T_j(x)$ in the exponent. $m_j(\eta)$ is that feature's average under the model, not an average of observed data.

3. Normalizer and moments

S3.05 · Slide 28/67

STUDY NOTE

Choosing natural coefficients determines a distribution, which in turn predicts feature averages. Those model averages are different objects from both the coefficients and the averages computed from observed data; fitting will connect them later.

YOUR NOTES

Lecture L02 — Exponential Families

Differentiating the mean gives a second derivative of A

Carried forward S3.04 · Established first-moment equality

$$\partial_j A(\eta) = E_\eta\left[T_j(X)\right] = \int T_j(x) p_\eta(x)\, \mathrm{d}\nu(x)$$

Updated: $$\partial_k \partial_j A(\eta) = \int T_j(x) \partial_k p_\eta(x)\, \mathrm{d}\nu(x)$$

Differentiate both sides, holding T fixed, under the same regularity condition

$$\partial_k \partial_j A(\eta) = \partial_k \int T_j(x) p_\eta(x)\, \mathrm{d}\nu(x)$$

3. Normalizer and moments

S3.06 · Slide 29/67

STUDY NOTE

The expected feature already equals a first derivative of A, so differentiating that equality produces a second derivative on one side. Inside the integral, the features remain fixed and only the density is differentiated, setting up the covariance calculation.

YOUR NOTES

Lecture L02 — Exponential Families

The density derivative compares a feature with its expectation

Carried forward S3.06 · Current integral; S1–S3 · Density and first moment

$$\partial_k \partial_j A = \int T_j(x) \partial_k p_\eta(x)\, \mathrm{d}\nu(x)$$

$$p_\eta(x) = h(x) e^{\eta^T T(x) - A(\eta)}, \quad \partial_k A = E_\eta[T_k(X)]$$

Updated: $\partial_k p_\eta(x) = p_\eta(x)\big(T_k(x) - E_\eta[T_k(X)]\big)$

Calculate the missing density derivative in the carried integral

$$\partial_k p_\eta(x) = h(x) e^{\eta^T T(x) - A(\eta)} \partial_k(\eta^T T(x) - A(\eta))$$
$$= p_\eta(x)(T_k(x) - \partial_k A(\eta))$$

3. Normalizer and moments

S3.07 · Slide 30/67

STUDY NOTE

Differentiating the exponent changes the observation weight according to how its feature differs from the model average. This centered feature expression is the missing ingredient needed to turn the derivative of the expected feature into a covariance.

YOUR NOTES

Lecture L02 — Exponential Families

Substituting that derivative exposes the covariance

Carried forward S3.06–07 · Current integral and computed density derivative

$$\partial_k \partial_j A = \int T_j(x) \partial_k p_\eta(x)\, \mathrm{d}\nu(x)$$

$$\partial_k p_\eta(x) = p_\eta(x)\big(T_k(x) - E_\eta[T_k]\big)$$

Updated: $\nabla^2 A(\eta) = \mathrm{Cov}_\eta[T(X)]$

Substitute the derivative; $E_\eta[T_k]$ is constant in x

$$\partial_k \partial_j A = \int T_j(x) p_\eta(x)\big(T_k(x) - E_\eta[T_k]\big)\, \mathrm{d}\nu(x)$$

$$= \int T_j T_k p_\eta \, \mathrm{d}\nu - E_\eta[T_k] \int T_j p_\eta \, \mathrm{d}\nu$$

$$= E_\eta\big[T_j T_k\big] - E_\eta\big[T_j\big] E_\eta[T_k] = \mathrm{Cov}_\eta\big(T_j, T_k\big)$$

3. Normalizer and moments

S3.08 · Slide 31/67

STUDY NOTE

The density derivative produces an expected product minus the product of expectations. Recognizing this covariance identifies the Hessian of A and explains how changes in natural coefficients affect the model's expected features.

YOUR NOTES

Lecture L02 — Exponential Families

Bernoulli checks both derivative identities

Carried forward S1 · Bernoulli coordinates; S3.04/08 · Moment identities

$T(x) = x, \quad A(\eta) = \log(1 + e^{\eta}), \quad \theta(\eta) = \frac{e^{\eta}}{1+e^{\eta}}$

$A'(\eta) = E_{\eta}[X], \quad A''(\eta) = \mathrm{Var}_{\eta}(X)$

$$A'(\eta) = \frac{e^{\eta}}{1 + e^{\eta}} = \theta(\eta) = E_{\eta}[X]$$

$$A''(\eta) = \frac{e^{\eta}(1 + e^{\eta}) - e^{2\eta}}{(1 + e^{\eta})^2}$$

$$= \frac{e^{\eta}}{(1 + e^{\eta})^2} = \theta(\eta)(1 - \theta(\eta)) = \mathrm{Var}_{\eta}(X)$$

3. Normalizer and moments

S3.09 · Slide 32/67

STUDY NOTE

Differentiating the Bernoulli log normalizer recovers its familiar mean and variance. This concrete check connects the general expectation and covariance identities to quantities already understood for a binary observation.

YOUR NOTES

Lecture L02 — Exponential Families

Section 3 summary: A records model moments and curvature

$$\nabla A(\eta) = E_\eta[T(X)]$$
$$\nabla^2 A(\eta) = \mathrm{Cov}_\eta[T(X)]$$
$$\text{Bernoulli:} \quad A' = \theta, \quad A'' = \theta(1-\theta)$$

TAKE HOME Differentiating the normalizer recovers expected features and their covariance. Natural coefficients weight features; mean parameters describe the model's expected features.

3. Normalizer and moments

S3.10 · Slide 33/67

STUDY NOTE

Normalization carries information beyond total probability: its logarithm determines expected features and their covariance through differentiation. These are model quantities at a chosen parameter, providing the moment side of the estimation argument.

YOUR NOTES

Section 4

Sample likelihood and moment matching

Summed statistics and qualified moment matching

YOUR NOTES

Lecture L02 — Exponential Families

Multiplying Bernoulli probabilities exposes a sample statistic

Carried forward S1 · One Bernoulli observation
$p_\eta(x) = e^{\eta x - A(\eta)}, \quad A(\eta) = \log(1 + e^\eta)$

$$D = (x_1, \ldots, x_N) \in \{0,1\}^N, \quad N \geq 1, \quad k \triangleq \sum_{n=1}^{N} x_n$$

$$p_\eta(D) = \prod_{n=1}^{N} p_\eta(x_n) = \prod_{n=1}^{N} e^{\eta x_n - A(\eta)}$$

$$= e^{\eta \sum_n x_n - NA(\eta)} = e^{\eta k - NA(\eta)}$$

- Assume the observations are conditionally IID given the parameter; D is one particular ordered binary vector.
- At fixed N, the parameter-dependent expression uses the data only through k.

4. Likelihood and fitting

S4.02 · Slide 35/67

STUDY NOTE

Conditional independence lets us multiply the observation probabilities and collect their exponents. For a fixed sample size, the number of successes carries all parameter dependence in this binary-sample likelihood, suggesting the general summed-statistic reduction.

YOUR NOTES

Lecture L02 — Exponential Families

The IID exponential-family likelihood uses summed features

Carried forward S1–S2 · One-observation EF form; S4.02 · IID sample

$p_\eta(x) = h(x)e^{\eta^T T(x) - A(\eta)}, \quad D = (x_1, ..., x_N)$ IID

Updated: $p_\eta(D) = H(D)e^{\eta^T S(D) - NA(\eta)}$

Multiply the one-observation probabilities and collect the data factors

$$S(D) \triangleq \sum_{n=1}^{N} T(x_n), \quad H(D) \triangleq \prod_{n=1}^{N} h(x_n)$$

$$p_\eta(D) = \prod_{n=1}^{N} h(x_n)e^{\eta^T T(x_n) - A(\eta)}$$

- IID licenses multiplication; $H(D)$ is fixed in η.

4. Likelihood and fitting

S4.03 · Slide 36/67

STUDY NOTE

Multiplying the common one-observation form accumulates features into a sample sum and base factors into a data-only multiplier. This factorization explains why the model can use a fixed-size statistic even when the dataset grows.

YOUR NOTES

Lecture L02 — Exponential Families

Sufficiency preserves information in the likelihood

Carried forward S4 · IID factorization

$p_\eta(D) = H(D)e^{\eta^T S(D) - NA(\eta)}, \quad S(D) = \sum_n T(x_n)$

$$S(D_1) = S(D_2), \quad N \text{ fixed}, \quad H(D_1), H(D_2) > 0$$

$$\Rightarrow \frac{p_\eta(D_1)}{p_\eta(D_2)} = \frac{H(D_1)}{H(D_2)}$$

- $S(D)$ is sufficient for η.
  - The dimension of $S(D)$ is fixed as N grows.
- Sufficiency is model dependent.
  - May not be sufficient under a different model.
  - May not be scientifically sufficient for a particular application.

STUDY NOTE

Datasets with the same statistic give likelihoods proportional by a factor that does not depend on the parameter. Sufficiency therefore preserves this model's parameter evidence, while leaving open whether that model or statistic serves a different scientific question.

YOUR NOTES

Lecture L02 — Exponential Families

The likelihood score compares observed and model features

Carried forward S4.03 · Complete likelihood; S3.04 · Model moment

$p_\eta(D) = H(D)e^{\eta^T S(D) - NA(\eta)}, \quad S(D) = \sum_n T(x_n)$

$\nabla A(\eta) = E_\eta[T(X)]$

Updated: $\nabla \ell(\eta; D) = S(D) - NE_\eta[T(X)]$

Take the log of the carried likelihood, then differentiate at fixed data

$$\ell(\eta; D) \triangleq \log p_\eta(D) = \eta^T S(D) - NA(\eta) + \sum_n \log h(x_n)$$

$$\nabla \ell(\eta; D) = S(D) - N\nabla A(\eta)$$

- Differentiate at fixed observed data with $H(D) > 0$. The score is the log-likelihood gradient; the data-only term has derivative zero.

4. Likelihood and fitting

S4.05 · Slide 38/67

STUDY NOTE

The log-likelihood gradient is the observed feature total minus the total predicted by the model. The normalizer derivative supplies the model expectation, making the score an interpretable discrepancy rather than an unrelated optimization formula.

YOUR NOTES

Lecture L02 — Exponential Families

A finite interior MLE matches model and sample averages

Carried forward S4.05 · Updated likelihood score

$\nabla \ell(\eta; D) = S(D) - N E_\eta[T(X)], \quad S(D) = \sum_{n=1}^{N} T(x_n)$

Updated: $E_{\hat{\eta}}[T(X)] = \frac{S(D)}{N} = \frac{1}{N} \sum_{n=1}^{N} T(x_n)$

Set the score to zero at an interior maximum-likelihood estimate

$0 = \nabla \ell(\hat{\eta}; D) = S(D) - N E_{\hat{\eta}}[T(X)]$

- A finite interior maximum-likelihood estimate $\hat{\eta}$ solves the score equation.
  - The fitted model's expected value equals the sample average for every feature.

4. Likelihood and fitting

S4.06 · Slide 39/67

STUDY NOTE

At a finite interior maximum, the score is zero, so the fitted model reproduces the observed feature averages. This is the moment-matching rule used in the applications; the interior condition matters because some valid probability estimates lie on a boundary.

YOUR NOTES

Lecture L02 — Exponential Families

Bernoulli separates interior estimates from boundary limits

Carried forward S4 · Finite-interior moment matching; S1/S3 · Bernoulli

$T(x) = x, \quad E_\eta[X] = \theta, \quad \eta = \log\left(\frac{\theta}{1-\theta}\right), \quad k = \sum_n x_n$

$E_{\hat{\eta}}[T(X)] = \frac{1}{N}\sum_n T(x_n)$ at a finite interior MLE

$$0 < k < N \quad \Rightarrow \quad \hat{\theta} = \frac{k}{N}, \quad \hat{\eta} = \log\left(\frac{k}{N-k}\right)$$

$$k = 0 \quad \Rightarrow \quad \hat{\theta} = 0, \quad \eta \to -\infty$$

$$k = N \quad \Rightarrow \quad \hat{\theta} = 1, \quad \eta \to +\infty$$

- All-failure and all-success samples have valid probability-space maxima. Those boundary distributions have no finite natural coordinate, so the interior equation has no finite solution there.

4. Likelihood and fitting

S4.07 · Slide 40/67

STUDY NOTE

A mixed binary sample gives finite log odds and the usual observed success proportion. All-success or all-failure samples still have probability-space maxima, but their natural coordinates are infinite limits, showing why the interior score equation needs qualification.

YOUR NOTES

Lecture L02 — Exponential Families

Summed features avoid rescanning the sample for every parameter

Carried forward S4 · Parameter-dependent likelihood and score
$\ell(\eta; D) = \eta^T S(D) - NA(\eta) + \text{data-only constant}$

$$D \quad \to \quad (N, S(D))$$

$$\eta^T S(D) - NA(\eta), \quad S(D) - N\nabla A(\eta)$$

- One pass accumulates $S(D)$: $O(Nd)$ work and $O(d)$ statistic storage.
- Each later parameter-dependent evaluation uses the stored sum: $O(d)$ arithmetic plus the cost of A or its gradient.
  - The reduction is exact for this family.
  - It does not validate the family or quantify parameter uncertainty.

4. Likelihood and fitting

S4.08 · Slide 41/67

STUDY NOTE

Once the sample statistic is accumulated, repeated likelihood and gradient calculations can reuse it instead of revisiting every observation. This is an exact computational saving within the chosen family, separate from assessing model adequacy or parameter uncertainty.

YOUR NOTES

Lecture L02 — Exponential Families

Section 4 summary: Summed statistics determine the likelihood

$$p_\eta(D) = H(D)e^{\eta^T S(D) - NA(\eta)}$$

$$E_{\hat{\eta}}[T(X)] = \frac{S(D)}{N} \quad \text{at a finite interior MLE}$$

TAKE HOME

Summed features preserve this family's likelihood information, with a fixed-size representation. At a finite interior maximum, model feature averages match sample averages. Boundary estimates may require infinite natural coordinates.

STUDY NOTE

The exponential-family form reduces IID learning to summed features and a model normalizer. At an existing finite interior maximum, the resulting score equation matches model and sample feature averages, with boundary examples marking the rule's limits.

YOUR NOTES

Section 5

Fit the number of recorded storms

Fit a recorded-event rate from five summers

YOUR NOTES

Lecture L02 — Exponential Families

One observation is a recorded event count per summer

Summer	Count n_y
1960	48
1961	48
1962	56
1963	37
1964	38

$$N_y \sim \text{Poisson}(m), \quad m > 0$$

$$p(n|m) = e^{-m}\frac{m^n}{n!}, \quad n \in \{0, 1, 2, ...\}$$

Original Illinois summer files, 1960–1964 · data/README.md

- m is an expected **recorded-event count per summer**, not a rainfall amount.
- The model assumes independent summers, a common rate and comparable recording effort.

STUDY NOTE

The modeled outcome is the number of recorded events in a summer, not the positive amount of rain in an event. A common-rate Poisson fit treats these five annual counts as comparable independent observations, connecting the general machinery to a specific data structure.

YOUR NOTES

Lecture L02 — Exponential Families

Collect the count and rate terms in the Poisson exponent

Carried forward S1–S2 · Exponential-family form; S5.02 · Poisson count model

$p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

$p(n|m) = e^{-m}\frac{m^n}{n!}, \quad m > 0, \quad n \in \{0, 1, 2, ...\}$

$$p(n|m) = \frac{1}{n!}\exp(n\log m - m)$$

$$\eta \triangleq \log m \Rightarrow m = e^\eta$$

$$p_\eta(n) = \frac{1}{n!}\exp(n\eta - e^\eta)$$

$$T(n) = n, \quad h(n) = \frac{1}{n!}, \quad A(\eta) = e^\eta$$

- Unlike Bernoulli, the fixed base factor varies with the outcome.

5. Poisson counts

S5.03 · Slide 45/67

STUDY NOTE

Taking the logarithm of the positive rate exposes the count as the sufficient statistic and the rate as the parameter-only exponential term. This matches the general form directly, while the factorial supplies a fixed base factor that varies with the outcome.

YOUR NOTES

Lecture L02 — Exponential Families

The Poisson mean and variance both equal the rate

Carried forward S5.03 · Poisson ingredients; S3.04/08 · Moments

$T(n) = n, \quad \eta = \log m, \quad A(\eta) = e^{\eta}$

$A'(\eta) = E_{\eta}[N], \quad A''(\eta) = \mathrm{Var}_{\eta}(N)$

$$E_{\eta}[N] = A'(\eta) = e^{\eta} = m$$

$$\mathrm{Var}_{\eta}(N) = A''(\eta) = e^{\eta} = m$$

5. Poisson counts

S5.04 · Slide 46/67

STUDY NOTE

The identified log normalizer can be differentiated immediately using the established moment identities. Equal mean and variance are therefore consequences of this Poisson model, providing a feature to compare with the observed annual counts after fitting.

YOUR NOTES

Lecture L02 — Exponential Families

Moment matching fits the count rate by the sample mean

Carried forward S4 · IID log likelihood; S5.03–04 · Poisson ingredients

$\ell(\eta; D) = \eta^T S(D) - N_{\text{obs}} A(\eta) + \sum_y \log h(n_y), \quad N_{\text{obs}} = 5$

$T(n) = n, \quad h(n) = (n!)^{-1}, \quad A(\eta) = e^\eta, \quad S(D) = \sum_{i=1}^5 T(x_i) = \sum_{i=1}^5 n_i$

$$\ell(\eta; D) = \eta \sum_{y=1}^{5} n_y - 5e^\eta - \sum_{y=1}^{5} \log(n_y!)$$

$$\frac{d\ell}{d\eta} = \sum_y n_y - 5e^\eta$$

$$\frac{d\ell}{d\eta} = 0 \Rightarrow \hat{m} = e^{\hat{\eta}} = \frac{1}{5} \sum_y n_y$$

- $\ell''(\eta) = -5e^\eta < 0$: a positive total gives a unique finite maximum.
- All-zero counts give $m = 0$, requiring $\eta \to -\infty$.

5. Poisson counts

S5.05 · Slide 47/67

STUDY NOTE

The general sample likelihood and moment equation now use the annual count totals. Matching the model's expected count to the observed average yields the rate estimate; the negative second derivative confirms the finite maximum when the total count is positive.

YOUR NOTES

Lecture L02 — Exponential Families

Five summers give a rate estimate and a limited dispersion check

Carried forward S5.02 · Recorded counts; S5.05 · Rate estimator
$(48, 48, 56, 37, 38)$, $\hat{m} = \overline{n}$

$$\hat{m} = \frac{227}{5} = 45.4 \text{ recorded events/summer}$$

$$s^2 = \frac{1}{4}\sum_{y=1}^{5}\left(n_y - 45.4\right)^2 = 62.8$$

$$\frac{s^2}{\overline{n}} = \frac{62.8}{45.4} \approx 1.38$$

- The Poisson model predicts variance 45.4 at the fitted rate; the observed sample variance is larger.
- This descriptive comparison neither validates nor decisively rejects equal dispersion, recording completeness or a stable event process.

STUDY NOTE

The annual counts estimate the expected recorded-event rate at 45.4 per summer. Their sample variance is somewhat larger than the fitted Poisson variance, but five observations and uncertain recording assumptions make this a descriptive comparison rather than decisive model validation or rejection.

YOUR NOTES

Lecture L02 — Exponential Families

Section 5 summary: Count, normalize and fit a recorded-event rate

$$p_\eta(n) = \tfrac{1}{n!}\exp(n\eta - e^\eta)$$
$$T(n) = n, \quad \eta = \log m, \quad A(\eta) = e^\eta$$
$$\hat{m} = \overline{n} = 45.4 \ \text{ recorded events/summer}$$

TAKE HOME

The statistic is the event count, and matching its expectation fits the rate. This is a model of the recorded-event process; its physical and recording assumptions remain unverified.

STUDY NOTE

The Poisson example reuses the same representation, derivative identities and moment-matching estimate on a new support. It fits the recorded count process under explicit assumptions; event amounts remain a different modeled quantity for the notebook.

YOUR NOTES

Section 6

Fit a categorical distribution

Fit one of several labels by counting

YOUR NOTES

Lecture L02 — Exponential Families

Indicator statistics encode one of several categories

$$p(x|\theta) = \theta_x, \quad x \in \{1, ..., K\}, \quad \theta_k = P(X = k)$$

$$\theta_k \geq 0, \quad \sum_{k=1}^{K} \theta_k = 1, \quad K \geq 2$$

$$T_k(x) \triangleq \mathbf{1}\{x = k\}, \quad k = 1, ..., K$$

$$K = 3, \quad x = 2 \Rightarrow T(x) = (0, 1, 0)^T$$

- One indicator equals 1; every other indicator equals 0. Their sample sums count the categories.

STUDY NOTE

Indicators translate a categorical outcome into features that can be counted across observations. Their expected values are category probabilities, making them the natural bridge between the exponential-family representation and frequency estimation.

YOUR NOTES

Lecture L02 — Exponential Families

Log ratios choose a baseline category

Carried forward S6.02 · Categorical probabilities

$p(x|\theta) = \theta_x, \quad x \in \{1, \ldots, K\}$

$\theta_k > 0, \quad \sum_{k=1}^{K} \theta_k = 1$

Updated: $p(x|\theta) = \theta_K e^{\eta_x}, \quad x = 1, \ldots, K$

Express each probability relative to category K

$$\eta_K \triangleq 0, \quad \eta_k \triangleq \log\left(\frac{\theta_k}{\theta_K}\right), \quad k < K$$

$$e^{\eta_k} = \frac{\theta_k}{\theta_K} \quad \Rightarrow \quad \theta_k = \theta_K e^{\eta_k}$$

6. Categorical fitting

S6.03 · Slide 52/67

STUDY NOTE

Positive category probabilities can be expressed relative to one chosen baseline. The resulting log ratios supply the natural coefficients, while the baseline choice leaves only the independent entries needed to specify the categorical distribution.

YOUR NOTES

Lecture L02 — Exponential Families

Normalizing exponentiated coordinates gives softmax probabilities

Carried forward S6.03 · Updated categorical distribution and log ratios

$p(x|\theta) = \theta_K e^{\eta_x}, \quad \eta_K = 0$

$\eta_k = \log\left(\frac{\theta_k}{\theta_K}\right) \quad (k < K), \quad \sum_k \theta_k = 1$

Updated: $p(x|\theta(\eta)) = \theta_x(\eta) = \dfrac{e^{\eta_x}}{Z(\eta)}$

Use the probabilities summing to one to determine the common factor

$$Z(\eta) \triangleq \sum_{j=1}^{K} e^{\eta_j}$$

$$1 = \sum_k \theta_k = \theta_K Z(\eta) \quad \Rightarrow \quad \theta_K = \frac{1}{Z(\eta)}$$

- **Softmax**: exponentiate each η_k, then divide by the sum of all the exponentials.

6. Categorical fitting

S6.04 · Slide 53/67

STUDY NOTE

The probabilities summing to one determines the common multiplier left by the log-ratio parameterization. Exponentiating each coefficient and dividing by the sum of exponentials gives softmax, a probability map whose parameters still need to be fitted from data.

YOUR NOTES

Lecture L02 — Exponential Families

The indicators select the observed category's coefficient

Carried forward S6.02/04 · Indicators and updated softmax; S1 · EF form

$p(x|\theta(\eta)) = \frac{e^{\eta_x}}{Z(\eta)}, \quad Z(\eta) = 1 + \sum_{k=1}^{K-1} e^{\eta_k}$

$T_k(x) = \mathbf{1}\{x = k\}, \quad \eta_K = 0$

$p_\eta(x) = h(x)\exp(\eta^T T(x) - A(\eta))$

Updated: $p(x|\theta(\eta)) = \exp(\eta^T T(x) - A(\eta))$

Match the selected coefficient to the exponential-family form

$$\eta = (\eta_1, \ldots, \eta_{K-1})^T, \quad T(x) = (T_1(x), \ldots, T_{K-1}(x))^T$$

$$\eta_x = \sum_{k=1}^{K-1} \eta_k T_k(x) = \eta^T T(x)$$

$$h(x) = 1, \quad A(\eta) = \log Z(\eta)$$

- We omit T_K because $\eta_K = 0$. At $x = K$, all retained indicators are zero.

6. Categorical fitting

S6.05 · Slide 54/67

STUDY NOTE

The retained indicators select the coefficient of the observed category, turning the categorical probability into an exponential-family inner product. Omitting the baseline indicator is valid because its coefficient is zero; a baseline observation consequently has every retained indicator equal to zero.

YOUR NOTES

Lecture L02 — Exponential Families

Matching indicator averages gives the observed frequencies

Carried forward S6.05 · Categorical EF; S4 · Finite-interior moment matching

$p(x|\theta(\eta)) = \exp(\eta^T T(x) - A(\eta)), \quad T_k(x) = \mathbf{1}\{x = k\}$

$E_{\hat{\eta}}[T_k(X)] = \frac{1}{N}\sum_{n=1}^{N} T_k(x_n), \quad k < K$

Updated: $p\left(x|\hat{\theta}\right) = \hat{\theta}_x = \frac{N_x}{N}$

Fit positive counts by matching the indicator means

$D = (x_1, ..., x_N)$ IID, $\quad N > 0, \quad N_k \triangleq \sum_n \mathbf{1}\{x_n = k\}$

$E_\eta[T_k(X)] = P(X = k|\theta(\eta)) = \theta_k, \quad k < K$

$\hat{\theta}_k = \frac{1}{N}\sum_{n=1}^{N} T_k(x_n) = \frac{N_k}{N}, \quad k < K$

- For category K, use $\sum_k \theta_k = 1$ and $\sum_k N_k = N$.

6. Categorical fitting

S6.06 · Slide 55/67

STUDY NOTE

For categorical indicators, the model expectation is the corresponding category probability and the empirical average is its observed frequency. The earlier moment-matching rule therefore gives the familiar count-and-normalize fit, with the final probability supplied by normalization.

YOUR NOTES

Lecture L02 — Exponential Families

A zero count puts the categorical MLE on the boundary

Carried forward S6.06 · Updated categorical fit; S6.04 · Finite softmax

$\hat{\theta}_k = \frac{N_k}{N}, \quad p\left(x|\hat{\theta}\right) = \frac{N_x}{N}$

$\theta_K(\eta) = \frac{1}{Z(\eta)} > 0 \quad \text{for finite} \quad \eta$

Updated: $p\left(1|\hat{\theta}\right) = \frac{2}{3}, \quad p\left(2|\hat{\theta}\right) = \frac{1}{3}, \quad p\left(3|\hat{\theta}\right) = 0$

Apply the frequency estimate to a sample with an unobserved category

$(N_1, N_2, N_3) = (2, 1, 0), \quad N = 3$

$\hat{\theta} = \left(\frac{2}{3}, \frac{1}{3}, 0\right)$

- Category 3 is in the sample space. Its zero fitted probability is a boundary estimate, represented by no finite softmax coordinates.

6. Categorical fitting

S6.07 · Slide 56/67

STUDY NOTE

A declared category can have a fitted probability of zero when it was unobserved. The frequency estimate remains valid on the probability simplex, although that boundary distribution cannot be represented by finite softmax coordinates.

YOUR NOTES

Lecture L02 — Exponential Families

Section 6 summary: Categorical learning is count and normalize

$$T_k(x) = \mathbf{1}\{x = k\}, \quad \eta_k = \log\left(\frac{\theta_k}{\theta_K}\right), \quad \eta_K = 0$$

$$p(x|\theta(\eta)) = \exp(\eta^T T(x) - A(\eta))$$

$$\theta_k = \frac{e^{\eta_k}}{\sum_j e^{\eta_j}}, \quad p\left(x|\hat{\theta}\right) = \frac{N_x}{N}$$

TAKE HOME

Log ratios parameterize positive category probabilities; normalized counts fit them. The baseline leaves K-1 free parameters. Zero counts can put the fitted probabilities on the boundary.

STUDY NOTE

Log ratios and softmax describe the categorical distribution, while the indicator moment equation fits it by observed frequencies. This separates parameterization from estimation and supplies the same categorical fit that will be reused for each observed context in the name model.

YOUR NOTES

Section 7

Fit categorical transitions to baby names

Fit, inspect, score and generate with categorical rows

YOUR NOTES

Lecture L02 — Exponential Families

Fit one categorical distribution for each preceding character

Carried forward S6 · Categorical distribution and frequency estimate

$p(x|\theta) = \theta_x, \quad \hat{\theta}_k = \frac{N_k}{\sum_l N_l}$

32,033 lowercase name records · letters a–z plus a period

$$p(k|j, P) = p(k|\theta) = \theta_k = P_{jk}, \quad P = (P_{jk})$$

$$P_{jk} \geq 0, \quad \sum_k P_{jk} = 1$$

The record `emma` becomes `.emma.`: a fixed start context, four letters, then termination.

Karpathy makemore names, pinned data · teacher worked-solution handoff

- j is the preceding character and k the next character. Each j selects one categorical distribution.

7. Baby-name transitions

S7.02 · Slide 59/67

STUDY NOTE

The preceding character selects which categorical probability vector to use for the next character. Each row of P stores one such vector, so fitting reuses the preceding section's frequency estimator; the new modeling choice is dependence on the observed context.

YOUR NOTES

Lecture L02 — Exponential Families

Count adjacent pairs within each name, including both boundaries

Carried forward S7.02 · Categorical row and record boundaries

$p(k|j,P) = P_{jk}, \quad \sum_k P_{jk} = 1$

Fixed start and END: ., emma → .emma.

Updated: $N_j \triangleq \sum_k N_{jk}, \quad L$ letters $\Rightarrow L+1$ transitions

Count only transitions within each boundary-marked record

$. \to e \to m \to m \to a \to .$

$N_{jk} \triangleq$ number of observed $j \to k$ pairs

$K = 27, \quad (N_{jk}) \in \mathbb{N}^{27\times 27}, \quad M = \sum_{\text{names}} (L+1) = 228146$

- For fixed j, N_{jk} is the category count and N_j counts all observations after j. Include start/END; never cross between names.

7. Baby-name transitions

S7.03 · Slide 60/67

STUDY NOTE

For a fixed preceding character, its transition counts are precisely the category counts needed by the familiar frequency estimator. Keeping each name's start and terminal pairs, and avoiding pairs across records, ensures these counts describe the declared name model.

YOUR NOTES

Lecture L02 — Exponential Families

Group the name likelihood by preceding character

Carried forward S7.02–03 · Categorical rows and within-name counts

$p(x|\theta) = \theta_x, \quad p(k|j,P) = P_{jk}, \quad \theta_k = P_{jk}$ for fixed j

$N_j = \sum_k N_{jk}$, fixed start and END: .

Updated: $\ell(P;D) = \sum_{j,k:N_{jk}>0} N_{jk} \log P_{jk}$

Multiply independent names and group by preceding character

$c_1 \ldots c_L$ is a name, $c_0 = c_{L+1} = .$

$$p(c_1 \ldots c_L \text{ and END} | P, c_0 = .) = \prod_{t=1}^{L+1} P_{c_{t-1},c_t}$$

$$\mathcal{L}(P;D) = \prod_j \underbrace{\prod_k P_{jk}^{N_{jk}}}_{\text{categorical likelihood after } j}$$

7. Baby-name transitions

S7.04 · Slide 61/67

STUDY NOTE

Multiplying the conditional probabilities of each complete name and grouping repeated transitions produces a categorical likelihood for every preceding character. This factorization licenses reuse of the categorical fit even though characters within a name are dependent under the model.

YOUR NOTES

Lecture L02 — Exponential Families

Apply the categorical frequency estimate for each preceding character

Carried forward S6 · Categorical fit; S7.03–04 · Counts and grouped likelihood

$p(k|j,P) = P_{jk}, \quad \ell(P;D) = \sum_{j,k:N_{jk}>0} N_{jk} \log P_{jk}$

Categorical fit: $\hat{\theta}_k = \frac{N_k}{\sum_l N_l}, \quad N_j = \sum_l N_{jl}$

Updated: $p\left(k|j,\hat{P}\right) = \hat{P}_{jk} = \dfrac{N_{jk}}{\sum_l N_{jl}} = \dfrac{N_{jk}}{N_j}, \quad N_j > 0$

Use the next-character counts after j

$\theta_k = P_{jk}$

Category counts: $\left(N_{j1}, \ldots, N_{jK}\right)$

$\hat{P}_{q,u} = \dfrac{206}{272} \approx 0.757$

After q ($N_q = 272$)

Next	Count	Probability
u	206	0.7574
END	28	0.1029
a	13	0.0478
i	13	0.0478
Other	12	0.0441

- Zero counts give 0, as in section 6. For $N_j = 0$, there is no data for that fit.

7. Baby-name transitions

S7.05 · Slide 62/67

STUDY NOTE

For a fixed preceding character, divide each next-character count by the total count following that character. This applies the same categorical frequency rule as before; a zero transition gets zero probability, whereas an entirely unobserved context has no data to determine its distribution.

YOUR NOTES

Lecture L02 — Exponential Families

A complete-name score includes its terminal transition

Carried forward S7.04–05 · Complete-name distribution and fitted rows

$p(c_1 \ldots c_L \text{ and END} | P, c_0 = .) = \prod_{t=1}^{L+1} P_{c_{t-1}, c_t}$

$\hat{P}_{jk} = \frac{N_{jk}}{N_j}, \quad N_j > 0, \quad c_{L+1} = .$

Updated: $\log p\Big(\text{emma and END} | \hat{P}, c_0 = .\Big) \approx -12.569 \text{ nats}$

$p\Big(\text{emma and END} | \hat{P}, c_0 = .\Big) \approx 3.478 \times 10^{-6}$

Multiply the fitted next-character probabilities along .emma.

$p\Big(\text{emma and END} | \hat{P}, c_0 = .\Big) = \hat{P}_{.,e} \hat{P}_{e,m} \hat{P}_{m,m} \hat{P}_{m,a} \hat{P}_{a,.}$

$c = (., e, m, m, a, .), \quad \log p\Big(\text{emma and END} | \hat{P}, c_0 = .\Big) = \sum_{t=1}^{5} \log \hat{P}_{c_{t-1}, c_t}$

- `b → f` was unobserved: `bf` has probability 0 and log score $-\infty$.

7. Baby-name transitions

S7.06 · Slide 63/67

STUDY NOTE

A complete-name probability multiplies the fitted categorical probabilities along the whole path from start through termination. Taking logarithms adds the corresponding log probabilities. A required zero-probability transition makes the name probability zero and its log score negative infinity.

YOUR NOTES

Lecture L02 — Exponential Families

Generate a name by repeated categorical draws

Carried forward S7.05 · Updated fitted next-character distribution

$p\left(k|j,\hat{P}\right) = \hat{P}_{jk} = \frac{N_{jk}}{N_j}, \quad N_j > 0$

$j = .$ at the start; sampling . terminates the name

1. Set the current context: $j = .$
2. Draw the next character using $\hat{\theta}_k = \hat{P}_{jk}$; stop if END (.) is drawn.
3. Otherwise append it, set j to that symbol, and draw again.

Unedited output · PCG64 seed 2147483647 · all 12 strings terminated

```
dulike    zoshan     car    ssloysthierana
alah      jamay      bey    m
aucerve   imieemmy   ciei   xavene
```

- A normalized fitted model can generate implausible strings: the preceding character alone discards longer spelling context.

7. Baby-name transitions

S7.07 · Slide 64/67

STUDY NOTE

Generation uses the same fitted categorical distributions that score names, selecting a new vector after each sampled character. The unedited examples illustrate what this model produces and why conditioning only on the preceding character can miss longer spelling patterns.

YOUR NOTES

Lecture L02 — Exponential Families

One counting pass supports fitting and repeated queries

Carried forward S7.04–05 · Grouped log likelihood and fitted probabilities

$\ell(P; D) = \sum_{j,k:N_{jk}>0} N_{jk} \log P_{jk}$

$p\left(k|j, \hat{P}\right) = \hat{P}_{jk} = \frac{N_{jk}}{N_j}, \quad N_j > 0$

Operation	Work or storage
Count within-name pairs	$O(M)$ for $M = 228146$ transitions
Store a dense row table	$O(K^2)$ with $K = 27$
Score one L-letter name	$O(L)$ factors, including END
Reevaluate grouped likelihood	$O(K^2)$ from the counts

- As in the earlier likelihood reduction, saved category counts replace repeated scans of the training strings.
- The file weights name records equally; it contains no birth frequencies. In-sample fitting and generated strings do not establish held-out accuracy.

STUDY NOTE

Stored transition counts support repeated grouped-likelihood evaluations without rescanning every training name. The resulting small computation is useful, but equally weighted name records, in-sample fitting and generated examples do not establish predictive performance on new data.

YOUR NOTES

Lecture L02 — Exponential Families

Section 7 summary: Categorical rows fit, score and generate names

$$p\left(k|j,\hat{P}\right) = \hat{P}_{jk} = \frac{N_{jk}}{\sum_k N_{jk}}$$

$$p\left(c_1 \ldots c_L \text{ and END}|\hat{P}, c_0 = .\right) = \prod_{t=1}^{L+1} \hat{P}_{c_{t-1},c_t}$$

TAKE HOME

The same categorical frequency estimate fits every preceding-character context. Products include start and termination; categorical draws generate names. Using only the preceding character ignores longer spelling context; generated examples do not establish held-out accuracy.

STUDY NOTE

One familiar categorical frequency estimate is reused for each preceding-character context. These fitted probabilities support complete-name scoring and generation, while the restricted context and absence of held-out evaluation limit what the demonstration establishes.

YOUR NOTES

Lecture L02 — Exponential Families

Notebook: Fit storm amounts and obtain summer return levels

From the lecture

$\hat{m} = 45.4$ recorded events/summer

Moment matching fits expected features to empirical averages.

In the notebook

$X > 0, \quad T(X) = (\log X, X)^T$

Fit a Gamma distribution to 227 recorded rainfall amounts in inches.

M is the summer maximum; $M = 0$ if there are no events.

$$P(M > r_T) = \frac{1}{T}, \quad T \in \{10, 20, 100\}$$

- Nine questions: fit the amount distribution, marginalize the unknown Poisson event count, then summarize the summer-maximum distribution.
- Fitted-parameter return levels are plug-in estimates; five summers cannot validate far-tail extrapolation.

Notebook

NB.01 · Slide 67/67

STUDY NOTE

The notebook fits a Gamma model to positive event amounts using the same feature-moment logic. Combining that fit with the Poisson count model requires marginalizing the unknown number of events before summarizing annual maxima by return levels; inserting fitted parameters does not account for their uncertainty.

YOUR NOTES